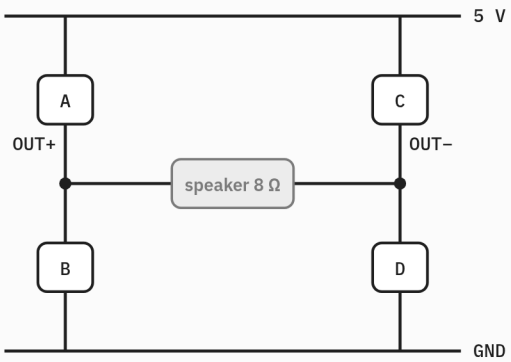
# The bridge

How the MAX98357A drives the speaker: four transistor switches in an H, the speaker as the crossbar, and only two legal ways to close them. Four drawings.

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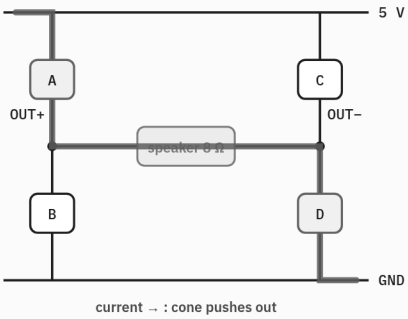
**The bridge: four switches and the speaker as the crossbar**




A, B, C, D are transistors inside the MAX98357A. The chip closes them in diagonal pairs only.

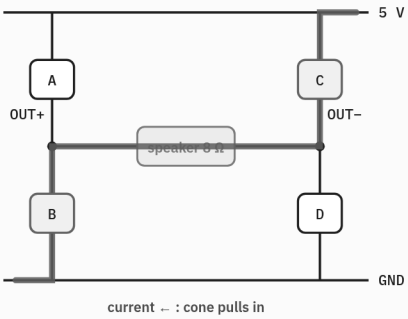
**It's an H.** Two legs from 5 V down to ground, two switches per leg, the speaker across the middle. OUT+ and OUT− are the midpoints of the legs: each one is either pulled up to 5 V by its top switch or down to 0 V by its bottom switch, never left floating.

**State 1: A + D closed**




current → : cone pushes out

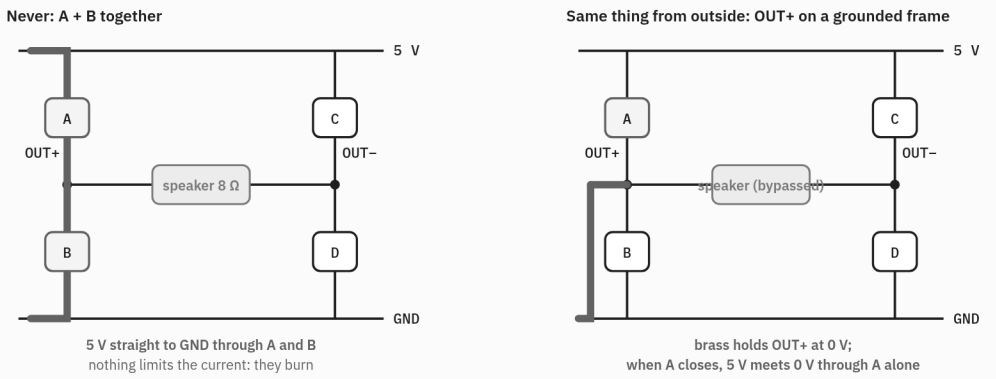
**State 2: C + B closed**




current ← : cone pulls in

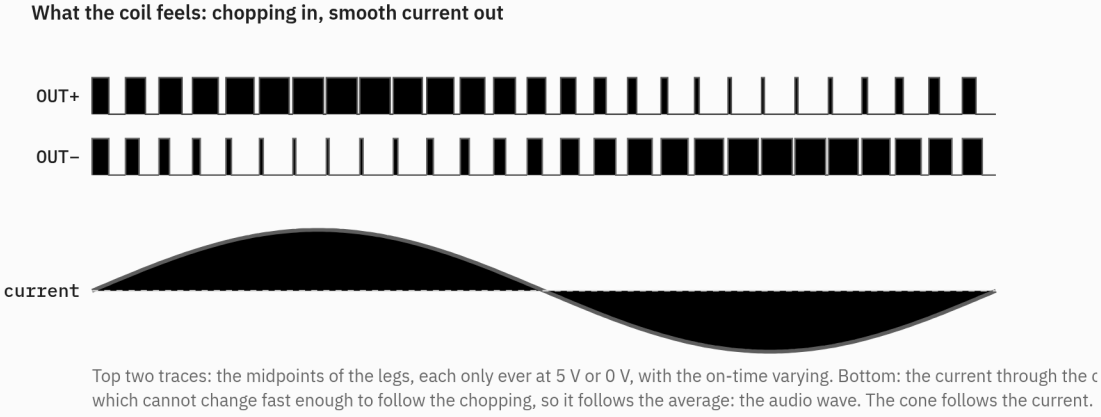
The chip flips between these two about 300 000 times a second. More time in state 1 = cone further out. More time in state 2 = further in. Equal = at rest.

**The same speaker, driven both ways from one supply.** That is the whole point of a bridge: 5 V across the speaker one way, then 5 V the other way, so the swing is twice what a single output could give and the power four times.




The chip never closes A with B on its own. A short to GND, a rail or a grounded frame does it for it.

**The one state that burns.** Both switches on one leg closed means 5 V to ground through nothing but a transistor. The chip's own logic forbids it; a wire from OUT+ or OUT− to any fixed voltage recreates it from outside. That is the whole reason those two pins go to the speaker and nowhere else.




Top two traces: the midpoints of the legs, each only ever at 5 V or 0 V, with the on-time varying. Bottom: the current through the c
which cannot change fast enough to follow the chopping, so it follows the average: the audio wave. The cone follows the current.

**Pulse width in, sound out.** This is where the speaker coil does the amplifier's last job for it: being an inductor, it averages the 300 kHz chopping into the smooth current the cone needs.

## In one breath

Two legs, four switches, diagonal pairs only. State 1 pushes the cone out, state 2 pulls it in, and the share of time in each is the sound. Both OUT pins are driven midpoints, so tying either to a fixed voltage forces a leg into the forbidden state. The speaker is the only thing that may sit between them.

## Parts

| PRODUCT YOU BOUGHT | HERE | LINK |
|---|---|---|
| HiLetgo MAX98357A amplifier | the four switches | B0CDWXZZCH |
| Same Sky CES-20134-088PM speaker, 8 Ω | the crossbar | DigiKey |

**Sources.** The bridge-tied-load output and the diagonal switching are the MAX98357A datasheet's architecture; the 300 kHz figure is the chip's class-D switching rate order of magnitude; the burn argument is Ohm's law applied to a transistor's on-resistance. The speaker-and-frame rules are the project's safety invariants in `CLAUDE.md`.

**What is and is not verified.** Conceptual drawings. The real chip adds dead time between switch states and current limiting that may save it from a brief short; do not count on it.